9.24 Solution: (a) Outside the sphere, assuming the conductivity is σ, the Maxwell equations become

$$\nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0, \quad \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J} = \sigma \mathbf{E}.$$

The wave equation is then

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} + \mu\epsilon \frac{\partial^2 \mathbf{E}}{\partial t^2} + \mu\sigma \frac{\partial \mathbf{E}}{\partial t} = 0.$$

Assuming harmonic time dependence, we can see that the characteristic equation is given by

$$\mu\epsilon\omega^2 + i\mu\sigma\omega - k^2 = 0,$$

compared to the corresponding equation is the free space, $\mu\epsilon\omega^2 - k^2 = 0$. Solution to the characteristic frequency in terms of k is

$$\omega = -i\frac{\sigma}{2\epsilon} \pm \sqrt{\frac{k^2}{\mu\epsilon} - \frac{\sigma^2}{4\epsilon^2}},$$

which always has a negative imaginary part.

The value of k can be determined in the same way as in Problem 9.22, but we need to consider TE and TM modes outside of the sphere, with $h_l^{(1)}(kr)$ replacing $j_l(kr)$. Therefore, for TE modes, the characteristic equation is

$$h_l^{(1)}(ka) = 0,$$

while for TM modes,

$$\frac{d}{dx}\left[x h_l^{(1)}(x)\right]\bigg|_{x=ka} = 0.$$